

Algebra II with Trigonometry

Chapter 6.5

Olympiad Solutions

Saved tutoring response with MathJax rendering. Original PDF file:
[4533652_alg-2trig-h-chp-6-5-note.pdf](#)

Solutions

1. Evaluate $\cos^{-1}(\sin(\sin^{-1} \frac{3}{5}))$ without a calculator, expressing your answer in radians.

Let $\theta = \sin^{-1} \frac{3}{5}$, so $\sin \theta = \frac{3}{5}$ and $\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$.

Now calculate:

$$\cos^{-1}(\sin(\sin^{-1} \frac{3}{5})) = \cos^{-1}\left(\frac{3}{5}\right)$$

But that's just $\cos^{-1}(\sin \theta)$.

Recall: For $\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$,

$$\cos^{-1}(\sin \theta) = \frac{\pi}{2} - \theta$$

because $\sin \theta = \cos(\frac{\pi}{2} - \theta)$, and $\cos^{-1}$ outputs values in $[0, \pi]$.

Therefore,

$$\cos^{-1}(\sin(\sin^{-1} \frac{3}{5})) = \frac{\pi}{2} - \sin^{-1} \frac{3}{5}$$

2. If $\tan^{-1}(x) + \tan^{-1}(y) = \frac{\pi}{4}$, what is the value of $x + y + xy$?

Let $A = \tan^{-1}(x)$, $B = \tan^{-1}(y)$. Given $A + B = \frac{\pi}{4}$.

Use the tangent addition formula:

$$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

Plugging in,

$$\tan \frac{\pi}{4} = 1 = \frac{x + y}{1 - xy}$$

Cross-multiplied:

$$1 - xy = x + y \implies x + y + xy = 1$$

Answer: 1

3. Simplify $\sin(\cos^{-1}(x)) + \cos(\sin^{-1}(x))$ **for** $x \in [-1, 1]$.

Let $\theta = \cos^{-1} x$, so $x = \cos \theta$, $\theta \in [0, \pi]$.

Then

$$\sin(\cos^{-1}(x)) = \sin(\theta) = \sqrt{1 - \cos^2 \theta} = \sqrt{1 - x^2}$$

(since $\sin \theta \geq 0$ on $[0, \pi]$).

Let $\phi = \sin^{-1} x$, $x = \sin \phi$, $\phi \in [-\frac{\pi}{2}, \frac{\pi}{2}]$, so

$$\cos(\sin^{-1}(x)) = \cos \phi = \sqrt{1 - x^2}$$

(because $\cos \phi \geq 0$ on this interval).

Sum:

$$\sin(\cos^{-1}(x)) + \cos(\sin^{-1}(x)) = \sqrt{1 - x^2} + \sqrt{1 - x^2} = 2\sqrt{1 - x^2}$$

Answer: $2\sqrt{1 - x^2}$

4. Without evaluating directly, decide if the following two are equal for $x \in [-\frac{\pi}{2}, \frac{\pi}{2}]$: $\arccos(\sin x)$ and $\arcsin(\cos x)$.

Let $y = x$, $x \in [-\frac{\pi}{2}, \frac{\pi}{2}]$.

First, note that:

$$\arccos(\sin x) = y_1, \quad \arcsin(\cos x) = y_2$$

Express both in terms of x :

- $\arcsin(\cos x)$: Since $\cos x \in [0, 1]$ in this interval, so value between 0 and 1,

$$\arcsin(\cos x) \in [0, \frac{\pi}{2}], \quad \text{and } \arcsin(\cos x) = \frac{\pi}{2} - x$$

(since $\sin(\frac{\pi}{2} - x) = \cos x$, and $\frac{\pi}{2} - x \in [0, \pi]$).

- $\arccos(\sin x)$: $\sin x \in [-1, 1]$, codomain for $\arccos$ is $[0, \pi]$.

Note that $\arccos(\sin x) = y \iff \cos y = \sin x$. Rearranged:
 $\sin x = \cos y \implies x = \frac{\pi}{2} - y$, or $y = \frac{\pi}{2} - x$.

But:

$$\arccos(\sin x) = \frac{\pi}{2} - x \quad \text{for } x \in [-\frac{\pi}{2}, \frac{\pi}{2}]$$

Same as above.

Conclusion: Yes, $\arccos(\sin x) = \arcsin(\cos x)$ for all $x \in [-\frac{\pi}{2}, \frac{\pi}{2}]$.

5. Find the exact value of $\tan(\arccos(\frac{1}{3}))$ in simplest radical form.

Let $\theta = \arccos(\frac{1}{3})$, so $\cos \theta = \frac{1}{3}$.

Draw right triangle: adjacent = 1, hypotenuse = 3, so opposite side by Pythagoras:

$$\text{opposite} = \sqrt{3^2 - 1^2} = \sqrt{9 - 1} = \sqrt{8} = 2\sqrt{2}$$

Thus,

$$\tan \theta = \frac{\text{opposite}}{\text{adjacent}} = \frac{2\sqrt{2}}{1}$$

Answer: $\boxed{2\sqrt{2}}$

6. Let $A = \arctan 2 + \arctan 3$. Find the exact value of $\tan A$.

Use tangent addition formula:

$$\tan(A) = \tan(\arctan 2 + \arctan 3) = \frac{2 + 3}{1 - (2)(3)} = \frac{5}{1 - 6} = \frac{5}{-5} = -1$$

Answer: $\boxed{-1}$